

A PROOF OF THEOREM 3.1

We prove the following precise form of the theorem. Recall from §3.2 that an arbitrary DNF predicate $\bigvee_j C_j$ with $C_j = \bigwedge_i p_{ji}$ is evaluated against the histograms atom by atom, so the operative guarantee is per-atom: *for each column (group) of $\mathcal{A}$, if $D(\mathbb{H}_i, \mathbb{H}_U) \leq \epsilon$ holds $\forall \mathcal{U}_i \subset \mathcal{U}$, then every predicate atom ϕ on that column satisfies $|\sigma_i(\phi) - \sigma_U(\phi)| \leq 2\epsilon$, where $\sigma_U(\phi)$ denotes the selectivity of ϕ within $\mathcal{U}$; for equality/IN columns the bound is moreover multiplicative; and, whenever $\sigma_U(\phi) > 2\epsilon$, the partition satisfies Proposition 1 with σ_{glob} read as $\sigma_U(\phi)$ and its constant instantiated as $2\epsilon/(\sigma_U(\phi) - 2\epsilon)$.*

Fix the column concerned by the atom ϕ and normalize both histograms to unit mass, so that $\mathbb{H}_i$ (resp., $\mathbb{H}_U$) represents the value distribution P_i (resp., P_U) of that column over $\mathcal{U}_i$ (resp., $\mathcal{U}$): for a value set S , $P(S)$ is the fraction of objects whose value falls in S . The atom selects a value set S_ϕ , with $\sigma_i(\phi) = P_i(S_\phi)$ and $\sigma_U(\phi) = P_U(S_\phi)$ (Equation (1)). We bound $|\sigma_i(\phi) - \sigma_U(\phi)|$ per column type.

Ordered columns. Here D is the KS distance of the normalized cumulative distributions $F_i(x) = P_i((-\infty, x])$ and $F_U(x) = P_U((-\infty, x])$, so the hypothesis reads $\sup_x |F_i(x) - F_U(x)| \leq \epsilon$. A range atom selects an interval $S_\phi = (a, b]$, hence

$$\begin{aligned} |\sigma_i(\phi) - \sigma_U(\phi)| &= |F_i(b) - F_i(a) - [F_U(b) - F_U(a)]| \\ &\leq |F_i(b) - F_U(b)| + |F_i(a) - F_U(a)| \leq 2\epsilon. \end{aligned}$$

A one-sided range ($a = -\infty$ or $b = +\infty$) drops one endpoint term and satisfies the sharper bound ϵ . The factor 2 is tight, as the two endpoint deviations may attain ϵ with opposite signs.

Equality/IN columns. Here every bucket is a single category v with masses $p_i(v)$ and $p_U(v)$, and the hypothesis reads $\max_v |p_i(v) - p_U(v)|/p_i(v) \leq \epsilon$, with the convention $0/0=0$ (a bucket with $p_i(v)=0$ satisfies this only if $p_U(v)=0$ as well, and then contributes nothing). An equality or IN atom selects a category set S_ϕ , hence

$$|\sigma_i(\phi) - \sigma_U(\phi)| \leq \sum_{v \in S_\phi} |p_i(v) - p_U(v)| \leq \epsilon \sum_{v \in S_\phi} p_i(v) = \epsilon \sigma_i(\phi) \leq \epsilon.$$

Since each bucket satisfies $(1 - \epsilon)p_i(v) \leq p_U(v) \leq (1 + \epsilon)p_i(v)$, summing over S_ϕ establishes the multiplicative form $(1 - \epsilon)\sigma_i(\phi) \leq \sigma_U(\phi) \leq (1 + \epsilon)\sigma_i(\phi)$.

In both cases $|\sigma_i(\phi) - \sigma_U(\phi)| \leq 2\epsilon$, which is the drift certified by the build-time discard test of §3.2.

From the drift bound to Proposition 1. Since $|\sigma_i(\phi) - \sigma_U(\phi)| \leq 2\epsilon$, whenever $\sigma_U(\phi) > 2\epsilon$,

$$\text{qerr}(\sigma_i(\phi), \sigma_U(\phi)) = \frac{\max(\sigma_i, \sigma_U)}{\min(\sigma_i, \sigma_U)} \leq 1 + \frac{|\sigma_i - \sigma_U|}{\min(\sigma_i, \sigma_U)} \leq 1 + \frac{2\epsilon}{\sigma_U(\phi) - 2\epsilon},$$

so the partition satisfies LSHomo with σ_{glob} read as $\sigma_U(\phi)$ and the constant of Proposition 1 instantiated as $2\epsilon/(\sigma_U(\phi) - 2\epsilon)$; for equality/IN columns the multiplicative form yields the sharper, σ_U -independent constant $\text{qerr} \leq 1/(1 - \epsilon) \leq 1 + 2\epsilon$ for $\epsilon \leq 1/2$, which the uniform constant above subsumes. For atoms with $\sigma_U(\phi) \leq 2\epsilon$ no multiplicative guarantee is claimed, but the additive bound—the form the cost model of §4.2 consumes—holds unconditionally.

Remark. A predicate over an ordered column whose value set is a union of k disjoint intervals is handled by the same endpoint argument, with bound $2k\epsilon$. Furthermore, the bound compares a

sub-region with its *tested* parent; along a chain of h consecutively discarded levels the per-level bounds telescope—to at most $2h\epsilon$ for ordered columns (triangle inequality for the KS distance) and to $(1 + \epsilon)^h - 1$, multiplicatively, for equality/IN columns—with respect to the histogram actually inherited, where h is bounded by the height of the forest.

Remark (noise-adjusted test at confidence $1 - \delta$). At build time the criterion is applied to the noise-adjusted discrepancy $\tilde{D}$ of §3.2 rather than to the raw empirical one: deviations within the DKW critical scale $t_\delta = s\sqrt{\frac{1}{2} \ln \frac{2}{\delta}}$, $s = \sqrt{1/n_i + 1/n_U}$, are attributed to sampling noise, so the premise $D \leq \epsilon$ of Theorem 3.1 is asserted for the underlying local distributions rather than for the raw empirical histograms. Dropping the correction ($t_\delta=0$) restores the raw deterministic test.

Confidence guarantee of the adjusted test. Suppose the sub-region inherits its parent’s distribution (the case the criterion should discard). For a numeric column, the two-sample DKW-type inequality [23, 36] gives $\Pr[D > t] \leq 2 \exp(-2t^2/s^2)$; the overlap between $\mathcal{U}_i$ and its parent is no complication—comparing $\mathcal{U}_i$ against its complement in $\mathcal{U}$ and rescaling only strengthens the exponential decay, so the stated bound holds a fortiori. Hence

$$\Pr[\tilde{D} > \epsilon] = \Pr[D > \epsilon + t_\delta] \leq \Pr[D > t_\delta] \leq 2 \exp(-2t_\delta^2/s^2) = \delta,$$

i.e., $\Pr[\tilde{D} \leq \epsilon] \geq 1 - \delta$, independently of n_i, n_U . For a categorical column, Hoeffding’s inequality [12] bounds each bucket’s deviation, $\Pr[|p_i(v) - p_U(v)| > t] \leq 2e^{-2t^2/s^2}$ (likewise a fortiori under overlap); inverting at level δ/W per bucket, i.e., $t_{\delta/W} = s\sqrt{\frac{1}{2} \ln \frac{2W}{\delta}}$, and taking a union bound over the W buckets, the event that every bucket’s deviation stays below its correction—and hence $\tilde{D} = 0 \leq \epsilon$ —holds with probability at least $1 - \delta$; the correction is inflated only by the factor $\sqrt{\ln(2W/\delta)/\ln(2/\delta)}$, growing only as $\sqrt{\ln W}$ with the number of buckets. The raw test ($t_\delta=0$) instead carries the n -dependent guarantee $1 - 2 \exp(-2\epsilon^2/s^2)$, vacuous once $s^2 \geq 2\epsilon^2/\ln 2$; the inversion thus converts a guarantee that vanishes on small regions into a uniform, distribution-free one. (A variant requiring neither sample independence nor any two-sample form follows from applying the one-sample DKW inequality [23] to each side at level $\delta/2$, enlarging the correction to $t_\delta = (\sqrt{1/n_i} + \sqrt{1/n_U})\sqrt{\frac{1}{2} \ln \frac{4}{\delta}}$ with the same guarantee.)

B PROOF OF THEOREM 4.1

(1) *Reachability:* the layer-0 graph is globally navigable, satisfying the standard HNSW connectivity and navigability assumptions. Navigation is predicate-independent (Algorithm 2 constrains admission, not navigation): a flood-typed region is swept over its entire member slice, independently of the graph inside the region, and hands one representative entry to every adjacent region from the build-time adjacency structure—a superset of the cross-region edges a point-level walk could discover; a best-first-typed region expands neighbors predicate-independently, with the second-phase diffusion maintaining its reachability closure. Consequently, the region-level frontier reaches every region graph-connected to the entry, without being truncated by the predicate.

(2) *No closer satisfying point is missed*: flood admission is unconditional over the full member slice—every member is predicate-checked, and any satisfying point closer to the query than the current threshold is admitted. Best-first employs early stopping, but its precondition (deterministic stopping at region boundaries combined with the second-phase cost-free diffusion) guarantees that no predicate-satisfying point closer than worst can lie in any pruned direction. Specifically, the boundary stop prevents overshooting beyond nearby candidates, while the second-phase diffusion restores reachability to regions that were temporarily concealed by early stopping.

(3) The value of worst is determined exclusively by predicate-satisfying points (Algorithm 2), so the pruning frontier is never prematurely tightened in a way that would exclude u^* . Taken together, these properties ensure that u^* is eventually visited and admitted while its distance still lies within the maintained threshold.

C PSEUDO-CODE OF INCREMENTAL MAINTENANCE

Algorithm 4 shows the detailed procedure for incremental maintenance *w.r.t.* Insert/Update/Delete/Rebuild.

Algorithm 4 Incremental maintenance

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1: procedure INSERT( $x$ )                                ▶ fully real-time
2:   HNSW-insert  $x$  (parent recognized as a byproduct, Algo-
3:     rithm 1 Phase A)
4:   ascend kept-histogram ancestors of  $x$  (incl. the root): bucket
5:     +1 per column
6:   ascend region-boundary ancestors ( $R^*$  up to the root): sub-
7:     tree count  $n + = 1$ 
8:   set presence bits at every ancestor, per column
9: procedure UPDATE( $x$ , col  $i$ , old  $\rightarrow$  new) ▶ purely batched-lazy
10:   $T \leftarrow \text{tree}(x)$                                 ▶ forest tree
11:   $T.\text{cnt\_mod} + = 1$ ; old value lingers in all histograms ▶ no
12:    bucket move
13:  set the new value's presence bit at every ancestor
14:  if  $T.\text{cnt\_mod} \geq \text{thresh\_mod}(T)$  then REBUILD( $T$ )
15: procedure DELETE( $x$ ) ▶ root counts real-time, the rest lazy
16:   $T \leftarrow \text{tree}(x)$                                 ▶ forest tree
17:  HNSW-tombstone  $x$  (search skips tombstones) ▶ presence
18:    bits stay set
19:  at  $x$ 's root ancestor only: bucket  $-1$  per column, subtree
20:    count  $n - = 1$ 
21:   $T.\text{cnt\_del} + = 1$ ; if  $T.\text{cnt\_del} \geq \text{thresh\_del}(T)$  then RE-
22:    BUILD( $T$ )
23: procedure REBUILD( $T$ )                                ▶ shared; isolated to tree  $T$ 
24:  rescan  $T$ 's current layer-0 points (incl. adds, excl. tomb-
25:    stones, updated values)
26:  re-run Phase B of Algorithm 1 within the tree's storage
27:    budget; refresh region-ids / histograms /  $n$ ; recompute presence
28:    bits; refresh membership slices and region adjacency
29:   $T.\text{cnt\_mod}, T.\text{cnt\_del} \leftarrow 0$                 ▶ other trees untouched

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